HOSPITAL CARDIOVASCULAR AI IMAGING CENTER

Epicardial & Pericoronary Fat AI Quantification Report

Automated 3D WholeBody SegResNet & CRISP-CT Coronary Inflammation Analysis

Patient ID: 0909
Exam Date: 2026-10-29
Modality: Cardiac CT (CCTA)

ELEVATED RISK

CRISP-CT Vascular Inflammation & Fat Burden: Elevated (~90th %ile)

Threshold: PVAT > -70.1 HU indicates active inflammation.

EPICARDIAL ADIPOSE TISSUE (EAT)

Total EAT Volume: 164.4 cm³

Mean Attenuation: -85.9 HU

Chamber Sub-Compartments:

- Left Ventricle (LV): 60.4 cm³
- Right Ventricle (RV): 71.6 cm³
- Left Atrium (LA): 36.2 cm³
- Right Atrium (RA): 30.6 cm³

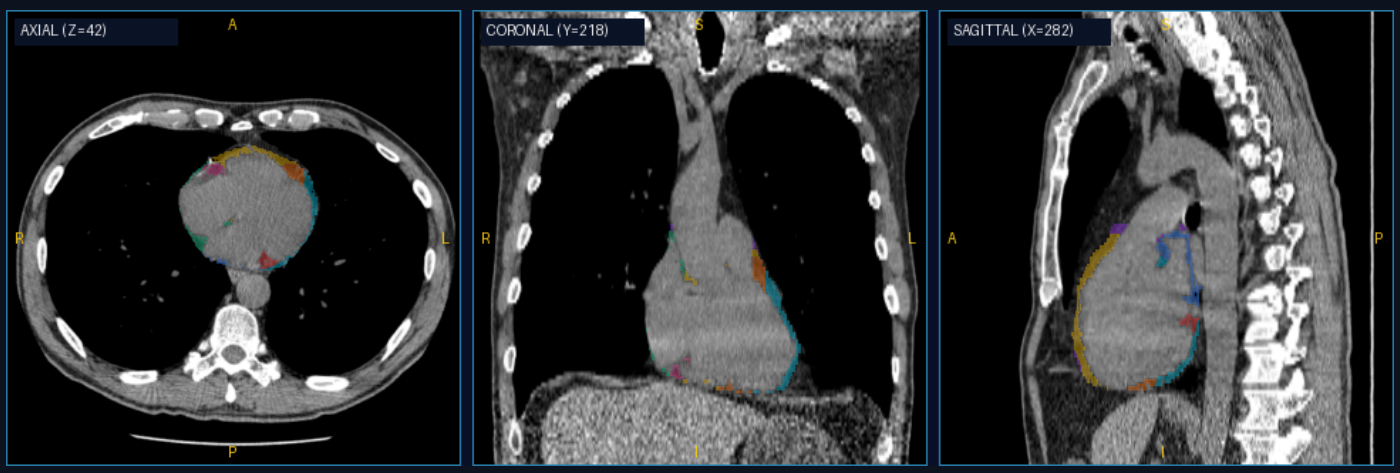
PERICORONARY ADIPOSE TISSUE (PVAT)

LAD PVAT:	14.9 cm³	-87.9 HU	Normal
LCX PVAT:	7.5 cm³	-83.2 HU	Normal
RCA PVAT:	10.1 cm³	-78.7 HU	Normal

Total Pericoronary Fat Volume: 32.5 cm³

3-VIEW ORTHOGONAL MULTI-PLANAR RECONSTRUCTION (MPR) WITH AI OVERLAYS

Axial (Z), Coronal (Y), and Sagittal (X) cardiac planes with 8-compartment adipose masks



STRUCTURED CLINICAL IMPRESSION (READY FOR EHR / POWERSCRIBE):

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QUANTITATIVE CARDIAC ADIPOSE & INFLAMMATION AI ANALYSIS:
- Overall Biomarker Triage: ELEVATED RISK
- Total Epicardial Fat (EAT) Volume: 164.4 cm3 (Elevated burden, ~90th %ile)
- Total EAT Mean Attenuation: -85.9 HU
- Pericoronary Fat Attenuation (CRISP-CT active inflammation threshold: > -70.1 HU):
    * LAD PVAT: -87.9 HU - Normal Attenuation (· -70.1 HU)
    * LCX PVAT: -83.2 HU - Normal Attenuation (· -70.1 HU)
    * RCA PVAT: -78.7 HU - Normal Attenuation (· -70.1 HU)
IMPRESSION: Total Epicardial Adipose Tissue (EAT) volume is 164.4 cm³ (Elevated burden, ~90th percentile), with mean attenuat
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For Investigational / Research Use Only (RUO). Clinical verification required by board-certified physician.

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AI GENERATIVE NARRATIVE FINDINGS & CLINICAL IMPRESSION:

TECHNIQUE: Non-contrast CT.

FINDINGS:
Medical devices: None present.

CHEST:
Lungs and airways: Trachea and both main bronchi are open. Centrilobular and paraseptal emphysema are observed in both lungs, more prominently at the apical levels. There are several tiny nonspecific nodules in both lungs.
Pleura: No pleural effusion or pneumothorax.
Mediastinum and hila: No enlarged lymph nodes in prevascular, pre-paratracheal, subcarinal, bilateral hilar, or axillary pathological dimensions.
Heart and pericardium: Normal heart size. No pericardial effusion or thickening. Calcific atheroma plaques are present in the coronary arteries.
Great vessels: Normal caliber thoracic aorta and main pulmonary artery.
Esophagus: Thoracic esophageal caliber is normal, and no significant tumoral wall thickening is detected.
Chest wall and breast: No chest wall or breast mass.
Bones: Bone structures in the study area are unremarkable. Vertebral body heights are preserved.

IMPRESSION:

1. Centrilobular and paraseptal emphysematous changes in both lungs.
2. Several tiny nonspecific nodules in both lungs.
3. Coronary atherosclerosis with significant epicardial adipose tissue and vessel-specific inflammatory changes.
<answer> Coronary artery wall calcification, Emphysema, Lung nodule </answer>

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